Amelia Taylor

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I'm an applied product data scientist who excels in deeply technical domains like biotech, streaming video technology, and logistics. I love helping technical teams make better decisions faster through metric development, experiment design and analysis, and causal inference. I balance statistical rigor with a practical focus on delivering useful solutions.

Technical Skills:
Languages & tools: Python (pandas, NumPy, scikit-learn), SQL, R (tidyverse), dbt, BigQuery, Git, Jupyter.
Visualization: Matplotlib, Altair, plotnine, Plotly, ggplot2.
AI workflows: AI-assisted development and agent skill development with Claude Code and Codex.

Experience:

- **Netflix** March 2024 - October 2026
  Senior Data Scientist | *Remote*

  – Developed streaming quality-of-experience (QoE) metrics, designed and analyzed experiments, and informed launch decisions in partnership with adaptive streaming engineers. Also supported experiment analysis and launch decisions across mobile, web, and TV player teams.

  – Designed and delivered a Cox proportional-hazards metric to evaluate streaming-quality trade-offs in live-event experiments, including statistical justification, stakeholder alignment, metric implementation, a supporting dbt table, and rollout.

  – Owned QoE measurement for Clips, establishing that an existing user-facing metric was sufficiently sensitive for early experimentation, identifying logging gaps and product bugs, and designing and building dedicated QoE tables.

  – Developed reusable quasi-experimental analysis methods and applied them in three live-streaming contexts, providing the evidence needed to proceed with rollout decisions.

  – Developed Python code and AI-agent skills for a Claude Code and Codex plugin that summarized experiment results using the same numbers and statistical inference as the experimentation platform. Within a day of rollout, two engineers used it to synthesize weeks of results and make a rollout decision in minutes.

- **Shopify** February 2022 - December 2023
  Staff Data Scientist | *Remote*

  Shopify is a commerce platform for merchants, making commerce better for everyone.

  – Led design and analysis of Shopify's one-page checkout experiment, addressing power and variance-estimation challenges with stratified merchant-level randomization and a generalized delta-method approach to checkout conversion. Built stakeholder confidence through simulation and partnered with engineers and product managers on implementation and restart decisions after an early bug. With two colleagues, delivered follow-up analyses across multiple segments under a tight turnaround, providing the confidence to roll out the product and pursue fix-forward experiments.

  – Drove improvements to the experimentation platform through collaboration with data scientists and the platform engineering team.

  – Advised teams across the organization on experiment design and analysis.

- **Zymergen, Inc** October 2016 - January 2022
  Staff Data Scientist (July 2019 - January 2022) | *Remote*
  Earlier roles: Data Scientist II (October 2016 - January 2019); Senior Data Scientist (January - July 2019); Data Science Manager (March - July 2019).

  Zymergen integrated robotics, software and genetic engineering for industrial fermentation.

  – Technical lead for two cross-functional teams delivering products combining statistics, chemical engineering and software engineering; standardized processes and reduced data processing time by at least two days per experiment.

  – Built an assay-development product that cut analytical time by 70% and made experimental design and analysis repeatable.

  - Developed and automated an outlier-detection method covered by a granted patent, saving two hours per week per client.
  - Designed and helped productionize normalized data tables that became company-wide infrastructure.
  - Worked with executive and director-level leadership to scope projects and assess ROI; coached data scientists taking on project management and technical leadership.

- **Selected Consulting** 2017 - 2018
  Consultant

  - *Mathematical Sciences Research Institute* (October 2017 - December 2018): evaluated a postdoctoral fellowship program using seven years of data, updated the survey, analyzed results in R and served as primary author of the final report.
  - *Insight Data Science* (September - November 2017): taught probability and mentored fellows in technical product development and interview preparation.

- **Academic Appointments**
  University Professor
  Colorado College: *Tenured Associate Professor*, Assistant Professor.
  Other appointments: Oregon State University–Cascades, St. Olaf College, and Rutgers University (Postdoctoral Fellow).

  - As department chair, supervised 11 full-time faculty, seven part-time faculty and two staff; coordinated departmental operations, personnel reviews and budgeting.
  - Developed a statistical method for phylogenetic inference using representation theory; implemented inference and simulation tests in R.
  - Converted statistical modeling and probability instruction to R; organized six week-long research workshops of more than 25 participants each.

Selected Presentations:

- **Data Did That!** Gordon Lecturer, Denison University October 2019
- **Data Did That!** National Mathematics Festival May 2019
- **Robots, Biology and Unsupervised Model Selection**, PyBay August 2018

Open Source Code Contributions:

- **statsmodels:** contributed two merged fixes to Student's quantile distribution (PR 7327, PR 7326).
- Proposed the Games–Howell statistical test (PR 7332) for statsmodels; in review.

Education (all degrees in Mathematics): Ph.D., *University of Kansas* | **M.S.**, *Purdue University* | **B.A.**, *St. Olaf College*